Area Between Curves

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area between a curve and the x-axis**

Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.

Roots at $x = \pm 2$. Area $= \int_{-2}^{2} (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

2 **[Foundational]** • **Target: Area between two curves (vertical strips)**

Find the area between $y = x^2$ and $y = x + 2$.

Intersect: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$: $x = -1, 2$. On this interval $x + 2 \geq x^2$. Area $= \int_{-1}^{2} [(x + 2) - x^2]dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{9}{2}$.

3 **[Foundational]** • **Target: Area between two curves (vertical strips)**

Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} \geq x^2$. Area $= \int_0^1 (\sqrt{x} - x^2)dx = \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

4 **[Foundational]** • **Target: Regions requiring horizontal strips**

Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y.

Intersect: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0$: $y = -1, 2$. On this interval $y + 2 \geq y^2$. Area $= \int_{-1}^{2} [(y + 2) - y^2]dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2} = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) = \frac{9}{2}$.

5 **[Foundational]** • **Target: Regions with multiple intersection points**

Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

On $[0, \frac{\pi}{4}]$, $\cos x \geq \sin x$; on $[\frac{\pi}{4}, \pi]$, $\sin x \geq \cos x$. Total area $= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi} (\sin x - \cos x)\,dx$. First: $[\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1$. Second: $[-\cos x - \sin x]_{\pi/4}^{\pi} = (1 - 0) - (-\sqrt{2}) = 1 + \sqrt{2}$. Total $= (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2} \approx 2.83$.

6 **[Foundational]** • **Target: Area using the more efficient variable of integration**

Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.

Solving for x: $x = 4 - y^2$, bounded by $x = 0$ (i.e. $y = \pm 2$). Integrating with respect to y is easier: Area $= \int_{-2}^{2} (4 - y^2)\,dy = \left[4y - \frac{y^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

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7 [Foundational] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.

On $[1, 2]$, $e^x \geq \ln x$ (since e^x grows much faster). Area $= \int_1^2 (e^x - \ln x)dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Setting up but not evaluating (AP-style)

Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

On $[-2, 0]$, $x^3 - 4x \geq 0$; on $[0, 2]$, $x^3 - 4x \leq 0$. Total area
$= \int_{-2}^{0} (x^3 - 4x)\,dx + \int_0^2 -(x^3 - 4x)\,dx = \int_{-2}^{0} (x^3 - 4x)\,dx - \int_0^2 (x^3 - 4x)\,dx$.

9 [Intermediate] • Target: Area between curves and a table of values

Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)]\,dx$.

Let $h(x) = f(x) - g(x)$: $h(0) = 7$, $h(2) = 4$, $h(4) = 4$, $h(6) = 6$. Trapezoidal Rule with width 2:
$T = 2\left[\frac{7+4}{2} + \frac{4+4}{2} + \frac{4+6}{2}\right] = 2[5.5 + 4 + 5] = 2(14.5) = 29$.

10 [Intermediate] • Target: Visualizing area between two curves

The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.

The line $y = x + 2$ lies above the parabola $y = x^2$ throughout $(-1, 2)$, matching the earlier setup $\int_{-1}^{2} [(x + 2) - x^2]dx = \frac{9}{2}$.

11 [Intermediate] • Target: A capstone area problem

Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \left|\int_0^1 (x^3 - x)\,dx\right|$.
$\int_{-1}^{0} (x^3 - x)\,dx = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} = 0 - \left(\frac{1}{4} - \frac{1}{2}\right) = \frac{1}{4}$. By symmetry the second piece is also $\frac{1}{4}$. Total area $= \frac{1}{2}$.

12 [Intermediate] • Target: Area between an exponential curve and a line

Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x, tangent only at $x = 0$).

Area $= \int_0^2 [e^x - (x + 1)]dx = \left[e^x - \frac{x^2}{2} - x\right]_0^2 = (e^2 - 2 - 2) - (1 - 0 - 0) = e^2 - 5$.

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13 **[Intermediate]** • **Target: An applied area-between-curves problem**

Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

Intersection: $50 - 2t = 30 + t \Rightarrow 20 = 3t \Rightarrow t = \frac{20}{3} \approx 6.67$. For $t < \frac{20}{3}$, $P_1 > P_2$; for $t > \frac{20}{3}$, $P_2 > P_1$. Total absolute difference $= \left| \int_0^{20/3} (P_1 - P_2)\,dt \right| + \left| \int_{20/3}^{10} (P_2 - P_1)\,dt \right|$, which can be computed piecewise using $P_1 - P_2 = 20 - 3t$.

14 **[Intermediate]** • **Target: Intersecting Curves with Multiple Regions**

Find the area between $y = x^3$ and $y = x^6$ on $[0, 1]$.

Final Answer: $\frac{3}{28}$. On $[0, 1]$, $x^3 \geq x^6$. Area $= \int_0^1 (x^3 - x^6)dx = \left[\frac{x^4}{4} - \frac{x^7}{7}\right]_0^1 = \frac{1}{4} - \frac{1}{7} = \frac{3}{28}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Area Partitions & Horizontal/Vertical Lines** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Find the area between $y = x$ and $y = x^3$ on $[0, 1]$.

Final Answer: $\frac{1}{4}$. On $[0, 1]$, $x \geq x^3$. Area $= \int_0^1 (x - x^3)dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

16 **[Challenging]** • **Target: Area between a curve and the x-axis** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Find the area between $y = x$ and $y = x^4$ on $[0, 1]$.

Final Answer: $\frac{3}{10}$. On $[0, 1]$, $x \geq x^4$. Area $= \int_0^1 (x - x^4)dx = \left[\frac{x^2}{2} - \frac{x^5}{5}\right]_0^1 = \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$.

17 **[Challenging]** • **Target: Area between two curves (vertical strips)** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Find the area between $y = x^4$ and $y = x^7$ on $[0, 1]$.

Final Answer: $\frac{3}{40}$. On $[0, 1]$, $x^4 \geq x^7$. Area $= \int_0^1 (x^4 - x^7)dx = \left[\frac{x^5}{5} - \frac{x^8}{8}\right]_0^1 = \frac{1}{5} - \frac{1}{8} = \frac{3}{40}$.

18 **[Challenging]** • **Target: Regions requiring horizontal strips** • **Cross-Lesson: 2.01 The Definition of the Derivative**

Find the area between $y = x^2$ and $y = x^5$ on $[0, 1]$.

Final Answer: $\frac{1}{6}$. On $[0, 1]$, $x^2 \geq x^5$. Area $= \int_0^1 (x^2 - x^5)dx = \left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$.

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19 [Challenging] • Target: Regions with multiple intersection points • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the total area between $y = x^3 - 36x$ and the x-axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.

Final Answer: 648. By symmetry, total area $= 2\int_0^6 (6^2x - x^3)dx = 2\left[\frac{6^2x^2}{2} - \frac{x^4}{4}\right]_0^6 = 2\left(\frac{1296}{2} - \frac{1296}{4}\right) = 2\left(\frac{1296}{4}\right) = \frac{1296}{2} = 648$.

20 [Challenging] • Target: Area using the more efficient variable of integration • Cross-Lesson: 3.01 The Chain Rule

Find the total area between $y = x^3 - 9x$ and the x-axis on $[-3, 3]$, accounting for the sign change at $x = -3, 0, 3$.

Final Answer: $\frac{81}{5}$. By symmetry, total area $= 2\int_0^3 (3^2x - x^3)dx = 2\left[\frac{3^2x^2}{2} - \frac{x^4}{4}\right]_0^3 = 2\left(\frac{81}{2} - \frac{81}{4}\right) = 2\left(\frac{81}{4}\right) = \frac{81}{2} = \frac{81}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Area between a curve and the x-axis

Find the area between $y = x^2$ and $y = -8x - 12$.

a. $\frac{35}{3}$

b. $\frac{32}{3}$ ✓

c. 8

d. $\frac{64}{3}$

Intersect: $x^2 = -8x - 12$ gives $x = -6, -2$. Area $= \int_{-6}^{-2} [(-8x - 12) - x^2]dx = \frac{(-2-(-6))^3}{6} = \frac{4^3}{6} = \frac{32}{3}$.

22 [Foundational] • Target: Area between two curves (vertical strips)

Find the area between $y = x^2$ and $y = 7x - 10$.

a. 9

b. 5

c. $\frac{9}{2}$ ✓

d. $\frac{11}{2}$

Intersect: $x^2 = 7x - 10$ gives $x = 2, 5$. Area $= \int_2^5 [(7x - 10) - x^2]dx = \frac{(5-(2))^3}{6} = \frac{3^3}{6} = \frac{9}{2}$.

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23 [Foundational] • Target: Regions requiring horizontal strips

Find the area between $y = x^2$ and $y = -x + 6$.

a. $\frac{25}{2}$

b. $\frac{131}{6}$

c. $\frac{125}{3}$

d. $\frac{125}{6}$ ✓

Intersect: $x^2 = -x + 6$ gives $x = -3, 2$. Area $= \int_{-3}^{2} [(-x + 6) - x^2] dx = \frac{(2-(-3))^3}{6} = \frac{5^3}{6} = \frac{125}{6}$.

24 [Foundational] • Target: Regions with multiple intersection points

Find the area between $y = x^2$ and $y = -2x$.

a. 2

b. $\frac{4}{3}$ ✓

c. $\frac{8}{3}$

d. $\frac{7}{3}$

Intersect: $x^2 = -2x$ gives $x = -2, 0$. Area $= \int_{-2}^{0} [(-2x) - x^2] dx = \frac{(0-(-2))^3}{6} = \frac{2^3}{6} = \frac{4}{3}$.

25 [Foundational] • Target: Area using the more efficient variable of integration

Find the area between $y = x^2$ and $y = 3x$.

a. 9

b. $\frac{11}{2}$

c. $\frac{9}{2}$ ✓

d. 4

Intersect: $x^2 = 3x$ gives $x = 0, 3$. Area $= \int_{0}^{3} [(3x) - x^2] dx = \frac{(3-(0))^3}{6} = \frac{3^3}{6} = \frac{9}{2}$.

26 [Foundational] • Target: Setting up but not evaluating (AP-style)

Find the area between $y = x^2$ and $y = 5x - 6$.

a. $\frac{1}{2}$

b. $\frac{1}{6}$ ✓

c. $\frac{1}{3}$

d. $\frac{7}{6}$

Intersect: $x^2 = 5x - 6$ gives $x = 2, 3$. Area $= \int_{2}^{3} [(5x - 6) - x^2] dx = \frac{(3-(2))^3}{6} = \frac{1^3}{6} = \frac{1}{6}$.

27 [Foundational] • Target: Area between curves and a table of values

Find the area between $y = x^2$ and $y = 6x$.

a. 72

b. 18

c. 37

d. 36 ✓

Intersect: $x^2 = 6x$ gives $x = 0, 6$. Area $= \int_{0}^{6} [(6x) - x^2] dx = \frac{(6-(0))^3}{6} = \frac{6^3}{6} = 36$.

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28 [Foundational] • Target: Visualizing area between two curves

Find the area between $y = x^2$ and $y = 9x - 14$.

a. $\frac{25}{2}$

b. $\frac{125}{6}$ ✓

c. $\frac{131}{6}$

d. $\frac{125}{3}$

Intersect: $x^2 = 9x - 14$ gives $x = 2, 7$. Area $= \int_2^7 [(9x - 14) - x^2]dx = \frac{(7-(2))^3}{6} = \frac{5^3}{6} = \frac{125}{6}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A capstone area problem

Find the area between $y = x^4$ and $y = x^9$ on $[0, 1]$.

a. $-\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$ ✓

d. $\frac{5}{36}$

On $[0, 1]$, $x^4 \geq x^9$. Area $= \int_0^1 (x^4 - x^9)dx = \left[\frac{x^5}{5} - \frac{x^{10}}{10}\right]_0^1 = \frac{1}{5} - \frac{1}{10} = \frac{1}{10}$.

30 [Intermediate] • Target: Area between an exponential curve and a line

Find the area between $y = x^2$ and $y = x^3$ on $[0, 1]$.

a. $\frac{1}{12}$ ✓

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $-\frac{1}{12}$

On $[0, 1]$, $x^2 \geq x^3$. Area $= \int_0^1 (x^2 - x^3)dx = \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

31 [Intermediate] • Target: An applied area-between-curves problem

Find the area between $y = x^3$ and $y = x^4$ on $[0, 1]$.

a. $\frac{9}{20}$

b. $-\frac{1}{20}$

c. $\frac{1}{12}$

d. $\frac{1}{20}$ ✓

On $[0, 1]$, $x^3 \geq x^4$. Area $= \int_0^1 (x^3 - x^4)dx = \left[\frac{x^4}{4} - \frac{x^5}{5}\right]_0^1 = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$.

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32 [Intermediate] • Target: Area Between Curves Integrated with Respect to x

Find the area between $y = x^4$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{30}$ ✓

b. $-\frac{1}{30}$

c. $\frac{1}{20}$

d. $\frac{11}{30}$

On $[0, 1]$, $x^4 \geq x^5$. Area $= \int_0^1 (x^4 - x^5)dx = \left[\frac{x^5}{5} - \frac{x^6}{6}\right]_0^1 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30}$.

33 [Intermediate] • Target: Area Between Curves Integrated with Respect to y

Find the area between $y = x^3$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{12}$ ✓

b. $\frac{5}{12}$

c. $\frac{2}{15}$

d. $-\frac{1}{12}$

On $[0, 1]$, $x^3 \geq x^5$. Area $= \int_0^1 (x^3 - x^5)dx = \left[\frac{x^4}{4} - \frac{x^6}{6}\right]_0^1 = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$.

34 [Intermediate] • Target: Intersecting Curves with Multiple Regions

Find the total area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the sign change at $x = -2, 0, 2$.

a. 4

b. $\frac{17}{2}$

c. 10

d. 8 ✓

By symmetry, total area $= 2\int_0^2 (2^2x - x^3)dx = 2\left[\frac{2^2x^2}{2} - \frac{x^4}{4}\right]_0^2 = 2\left(\frac{16}{2} - \frac{16}{4}\right) = 2\left(\frac{16}{4}\right) = \frac{16}{2} = 8$.

35 [Intermediate] • Target: Area Partitions & Horizontal/Vertical Lines

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

a. 39

b. 29

c. 31

d. 30 ✓

$T = 1\left[\frac{8+15}{2} + \frac{15+6}{2} + \frac{6+10}{2}\right] = 1\left[\frac{8+10}{2} + 15 + 6\right] = 1(9 + 15 + 6) = 30$.

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36 [Intermediate] • Target: Area between a curve and the x-axis

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 6]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 2$, $h(2) = 12$, $h(4) = 13$, $h(6) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 2) to approximate the area between the curves.

a. 58

b. 56

c. 52

d. 54 ✓

$T = 2\left[\frac{2+12}{2} + \frac{12+13}{2} + \frac{13+2}{2}\right] = 2\left[\frac{2+2}{2} + 12 + 13\right] = 2(2 + 12 + 13) = 54.$

37 [Intermediate] • Target: Area between two curves (vertical strips)

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 10$, $h(4) = 15$, $h(8) = 8$, $h(12) = 8$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.

a. 128 ✓

b. 132

c. 124

d. 164

$T = 4\left[\frac{10+15}{2} + \frac{15+8}{2} + \frac{8+8}{2}\right] = 4\left[\frac{10+8}{2} + 15 + 8\right] = 4(9 + 15 + 8) = 128.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Regions requiring horizontal strips • Cross-Lesson: 2.01 The Definition of the Derivative

Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 2]$ (note $e^{2x} \geq 2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 21.8

b. 24.96

c. 16.64

d. 20.8 ✓

$\text{Area } = \int_0^2 [e^{2x} - (2x + 1)]dx = \left[\frac{e^{2x}}{2} - \frac{2x^2}{2} - x\right]_0^2 \approx 20.8.$

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39 [Challenging] • Target: Regions with multiple intersection points • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 1]$ (note $e^{2x} \geq 2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 2.19

b. 1.43

c. 0.96

d. 1.19 ✓

Area $= \int_0^1 [e^{2x} - (2x + 1)]dx = \left[\frac{e^{2x}}{2} - \frac{2x^2}{2} - x\right]_0^1 \approx 1.19$.

40 [Challenging] • Target: Area Between Curves Integrated with Respect to x • Cross-Lesson: 3.01 The Chain Rule

What is the area of the region enclosed by $y = x^2$ and $y = 4$?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 16

d. 8

Intersection points: $x^2 = 4 \Rightarrow x = \pm 2$. Area $= \int_{-2}^{2}(4 - x^2)dx = 2[4x - \frac{x^3}{3}]_0^2 = 2(8 - \frac{8}{3}) = 2(\frac{16}{3}) = \frac{32}{3}$.

41 [Challenging] • Target: Area Between Curves Integrated with Respect to x • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Which integral represents the area enclosed by $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{4}$?

a. $\int_0^{\pi/4} (\cos x - \sin x)dx$ ✓

b. $\int_0^{\pi/4} (\sin x - \cos x)dx$

c. $\int_0^{\pi/4} (\cos x + \sin x)dx$

d. $\int_0^{\pi/2} \cos x dx$

On $[0, \pi/4]$, $\cos x \geq \sin x$. The upper curve is $\cos x$ and lower is $\sin x$, so $\int_0^{\pi/4} (\cos x - \sin x)dx$.

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42 [Challenging] • Target: Area Between Curves Integrated with Respect to y • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

When is it advantageous to integrate with respect to y rather than x?

a. When the boundary curves are expressed as $x = f(y)$ and solving for y requires multiple piecewise branches ✓

b. When the area is negative

c. Always for quadratic curves

d. Never on the AP Exam

Integrating with respect to y (right curve minus left curve) simplifies the setup whenever solving for y yields plus/minus roots and splits the region into multiple integrals.

43 [Challenging] • Target: Intersecting Curves with Multiple Regions • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The graphs of $y = x^3$ and $y = x$ enclose how many symmetric regions?

a. 2 symmetric regions ✓

b. 1 region

c. 3 regions

d. Infinitely many

Intersections: $x^3 - x = 0 \Rightarrow x(x-1)(x+1) = 0 \Rightarrow x = -1, 0, 1$. The curves enclose 2 symmetric regions (on $[-1, 0]$ and $[0, 1]$).

44 [Challenging] • Target: Intersecting Curves with Multiple Regions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Why does $\int_{-1}^{1}(x - x^3)dx = 0$ NOT represent the enclosed area?

a. The integrand is odd, causing positive and negative signed areas to cancel out ✓

b. The curves do not intersect at 0

c. The integral diverges

d. The bounds are incorrect

A definite integral computes net signed area. For true area, one must integrate $|f(x) - g(x)|$.

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45 [Challenging] • Target: Area Partitions & Horizontal/Vertical Lines • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find the area enclosed by $y = e^x$, $y = e^{-x}$, and the vertical line $x = 1$.

a. $e + \frac{1}{e} - 2$ ✓

b. $e - 1$

c. $e + \frac{1}{e}$

d. $2e - 2$

Intersections: $e^x = e^{-x} \Rightarrow x = 0$. On $[0, 1]$, $e^x \geq e^{-x}$. Area $= \int_0^1 (e^x - e^{-x})dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2$.

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